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Academic year 2024-2025 (Odd Sem)

Date	27/01/2025	Maximum Marks	50+10
Course Code	IS353IA	Duration	90+30
Sem	V	CIE III (Scheme)	
UG/~~PG~~	UG	Faculty: MEM/AS/VH/VG/JS/SHRS/ARA	

Artificial Intelligence and Machine Learning
(Common to AIML/CSE/CD/CY/ISE)

Note: - Students need to add comments to their answers wherever required.

Q. No.	Questions	M
1. A	The **utility function** represents the payoff each player receives in a game. UTILITY(s, p) A game is called a **zero-sum game** because the total payoff is constant (usually zero), meaning one player's gain is exactly balanced by the other player's loss.	**5**
B	Alpha-Beta Pruning is an optimization technique for the Minimax algorithm. It reduces the number of nodes evaluated in the search tree by eliminating branches that cannot affect the final decision. The algorithm uses two parameters: • **Alpha (α):** The best value that the maximizing player is guaranteed. • **Beta (β):** The best value that the minimizing player is guaranteed. **Efficiency of Alpha-Beta Pruning** 1. **Reduction in Nodes:** The number of nodes explored is significantly reduced, especially in deeper trees. o Without pruning: $O(b^d)$, where b is the branching factor, and d is the depth. o With pruning: Best-case $O(b^{d/2})$. 2. **Preserves Correctness:** The same optimal decision as Minimax is guaranteed. 3. **Improves Practicality:** Pruning allows deeper exploration within the same time constraints. By intelligently skipping unnecessary branches, Alpha-Beta Pruning ensures faster decision-making while maintaining the integrity of the Minimax result.	**5**
2. A	Generalization error Gen.error(m) = c_1 * Noise + Bias (m) + c_2 * Variance (m) Ensemble Method increase classification accuracy: Rational for ensemble method $$e_{\text{ensemble}} = \sum_{i=13}^{25} \binom{25}{i} \epsilon^i (1-\epsilon)^{25-i} = 0.06,$$	**05**
B	**Bagging** (Bootstrap Aggregating) is an ensemble learning technique that improves model accuracy by training multiple models on different random subsets of the data (created	**05**

through sampling with replacement) and then aggregating their predictions. For classification, it uses majority voting, and for regression, it averages the predictions. Bagging reduces variance and helps prevent overfitting,

3. A

Algorithm 6.6 AdaBoost algorithm.

1: $\mathbf{w} = \{w_j = 1/N \mid j = 1, 2, \ldots, N\}$. {Initialize the weights for all N examples.}
2: Let k be the number of boosting rounds.
3: **for** $i = 1$ to k **do**
4: Create training set D_i by sampling (with replacement) from D according to $\mathbf{w}$.
5: Train a base classifier C_i on D_i.
6: Apply C_i to all examples in the original training set, D.
7: $\epsilon_i = \frac{1}{N}\left[\sum_j w_j\, \delta(C_i(x_j) \neq y_j)\right]$ {Calculate the weighted error.}
8: **if** $\epsilon_i > 0.5$ **then**
9: $\mathbf{w} = \{w_j = 1/N \mid j = 1, 2, \ldots, N\}$. {Reset the weights for all N examples.}
10: Go back to Step 4.
11: **end if**
12: $\alpha_i = \frac{1}{2} \ln \frac{1-\epsilon_i}{\epsilon_i}$.
13: Update the weight of each example according to Equation 6.103.
14: **end for**
15: $C^*(\mathbf{x}) = \underset{y}{\operatorname{argmax}} \sum_{j=1}^{T} \alpha_j \delta(C_j(\mathbf{x}) = y))$.

10

Adjust weights

Error rate

$$\epsilon_i = \frac{1}{N}\left[\sum_{j=1}^{N} w_j\ I\left(C_i(\mathbf{x}_j) \neq y_j\right)\right],$$

Importance of classifier

$$\alpha_i = \frac{1}{2} \ln\left(\frac{1-\epsilon_i}{\epsilon_i}\right).$$

Weight adjustment

$$w_i^{(j+1)} = \frac{w_i^{(j)}}{Z_j} \times \begin{cases} e^{-\alpha_j} & \text{if } C_j(\mathbf{x_i}) = y_i \\ e^{\alpha_j} & \text{if } C_j(\mathbf{x_i}) \neq y_i \end{cases},$$

4. A

Instance	Actual Label	Majority Vote	Correct ?
1	1	1	1
2	0	0	1
3	1	1	0

5

4	0	0	0		
5	1	1	1		

Ensemble Accuracy = Correct Predictions/Total Instances=5/5=100%

B — **Cluster Analysis?** — 5

Cluster analysis is a technique to group data points into clusters where points in the same cluster are similar, and points in different clusters are dissimilar.

Types of Clustering Techniques

1. **Partitioning Clustering**: Divides data into non-overlapping clusters (e.g., K-means).
2. **Hierarchical Clustering**: Builds a tree of clusters using agglomerative or divisive approaches.
3. **Density-Based Clustering**: Forms clusters based on dense regions of data (e.g., DBSCAN).
4. **Grid-Based Clustering**: Divides data into grids and clusters cells with dense data (e.g., STING).

5. A — **Steps of the Algorithm:** — **05**

1. **Initialize Centroids**: Randomly choose k initial cluster centroids.
2. **Assign Points to Clusters**: Assign each data point to the nearest centroid.
3. **Update Centroids**: Recalculate centroids as the mean of all points in a cluster.
4. **Repeat**: Reassign points and update centroids until convergence (no changes in cluster assignments).

One Numerical Example

B — **05**

1. Cohesion (Intra-cluster Compactness)

- **Definition**: Measures how closely related the data points within the same cluster are.
- **Formula**: Cohesion = $\sum_{i=1}^{k} \sum_{x \in C_i} distance\ (x, centroid\ (C_i))$
- **Goal**: Minimize cohesion to ensure data points within a cluster are similar.

2. Separation (Inter-cluster Distinction)

- **Definition**: Measures how well-separated different clusters are from each other.
- **Formula**: Separation = $\sum_{i=1}^{k} \sum_{j=i+1}^{k} distance(centroid\ of\ C_i\ , centroid\ of\ C_j)$
- **Goal**: Maximize separation to ensure clusters are distinct from one another.

Combined Evaluation

The quality of clustering improves when cohesion is minimized, and separation is maximized. Metrics like the **Silhouette Coefficient** combine these concepts for practical evaluation.

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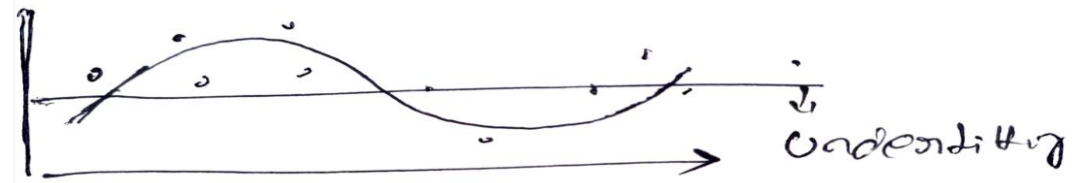
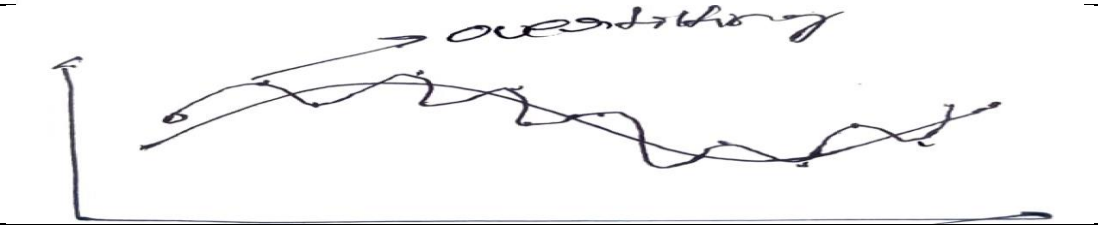
QUIZ

Q. No.	Questions	M
1.		2
2.		2
3.	Cohesion (C_i) = $\sum_{\substack{x \in C_i \\ y \in C_i}} proximity\ (x, y)$ Separation (C_i, C_j) = $\sum_{\substack{x \in C_i \\ y \in C_j}} proximity\ (x, y)$	2
4.	Parallel, Serial	2
5.	The performance of Random Forests is reduced as the fraction of irrelevant features increases because the irrelevant features introduce noise and reduce the quality of splits in the decision trees.	2

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